

Git & GitHub: Step-by-Step Guide

1. Introduction to Git and GitHub

- **Git:** A version control system to track changes in your code.
 - **GitHub:** A cloud platform to host Git repositories.
 - **Difference:** Git is local; GitHub is remote and collaborative.
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2. Setting up Git

- **Install Git:** Download from <https://git-scm.com/>.
- **Configure Git:**

```
git config --global user.name "Your Name"  
git config --global user.email "you@example.com"
```

- **Check version:**

```
git --version
```

3. Creating a Repository

- **Initialize local repo:**

```
git init
```

- **Create GitHub repo:** Go to GitHub > New Repository.
- **Connect local repo to GitHub:**

```
git remote add origin https://github.com/<USERNAME>/<REPOSITORY>.git
```

4. Basic Git Commands

- **Check status:**

```
git status
```

- **Stage files:**

```
git add .
```

- Commit changes:

```
git commit -m "Initial commit"
```

- View commit history:

```
git log
```

5. Working with Branches

- Create new branch:

```
git checkout -b new-branch
```

- Switch branch:

```
git checkout branch-name
```

- Merge branch:

```
git checkout main  
git merge new-branch
```

- Delete branch:

```
git branch -d branch-name
```

6. Pushing & Pulling

- Push to GitHub:

```
git push
```

- Push new branch:

```
git push -u origin new-branch
```

- Pull from GitHub:

```
git pull
```

7. Handling Files

- Ignore files (.gitignore):

```
node_modules/
```

- Remove sensitive file from repo:

```
git rm --cached .env  
git commit -m "Remove .env"  
git push
```

- Add to future ignores:

```
echo ".env" >> .gitignore  
git add .gitignore  
git commit -m "Ignore .env"  
git push
```

8. Cloning Repositories

```
git clone https://github.com/<USERNAME>/<REPOSITORY>.git
```

- Work on cloned repo by creating branches, committing, and pushing.

9. Undoing Changes

- Discard changes in a file:

```
git checkout -- filename
```

- Revert a commit:

```
git revert <commit-hash>
```

- Reset last commit locally:

```
git reset --soft HEAD~1
```

10. Advanced Tips

- View changes:

```
git diff
```

- Stash changes:

```
git stash  
git stash pop
```

- Resolve merge conflicts manually in files and commit.
 - Pull requests: Use GitHub to review and merge branches.
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11. Best Practices

- Use meaningful commit messages.
 - Use branches for features/bug fixes.
 - Never commit secrets; use .gitignore.
 - Regularly pull updates from remote repo.
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End of Guide